# REPURPOSING FDA-APPROVED ZINC CHELATING DRUGS AS INHIBITORS OF NEW DELHI METALLO-β-LACTAMASE-1 (NDM-1)

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## ABSTRACT

The erupt of broad spectrum antibiotic resistance due to NDM-1 beta-lactamase has become the perilous problem globally. This enzyme requires zinc ions at its catalytic site to hydrolyze beta-lactam antibiotics leading to antibiotic resistance towards multiple drugs. Targeting the zinc chelation in the enzyme offers pharmacological suppression. In this study an in silico drug repurposing approach was used to evaluate zinc coordinating compounds as enzymatic blockade of NDM-1. Four compounds as enzymatic blockade are used—Trientine, putrescine, spermine and spermidine. These were modeled via in silico docking to assess their binding affinity with NDM-1 enzyme. The results showed electrostatic complementarity with active site residues suggesting possible interference with zinc binding. ADME results evaluated the pharmacokinetic properties of these four compounds. Among the positive controls and the experimental drug , the experimental drug trientine exhibits therapeutic potential. These findings show the drug likeness of zinc chelating compounds as promising pharmacological inhibitors of NDM-1 enzymes.

## INTRODUCTION

Anti microbial resistance caused by NDM-1-beta lactamase is a serious health threat worldwide. This enzyme uses zinc ions present in the catalytic site to inactivate the beta-lactam antibiotics by hydrolyzing the beta-lactam rings including carbapenems. Targeting zinc chelation through metal chelating compounds and repurposing a drug is a promising master plan to inhibit NDM-1 activity. In this current study we repurposed the drug trientine by molecular docking and ADME analysis.

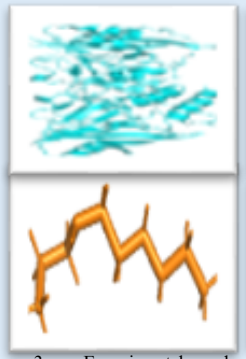
## OBJECTIVES

1. To predict the binding affinity of Trientine (Experimental ligand) and compare it with the positive controls (putrescine, spermidine and spermine) and for repurposing the zinc chelating drug using molecular docking.
2. To establish pharmacokinetic properties and drug-likeness of the experimental drug through ADME analysis for stating trientine as potential drug inhibitor.

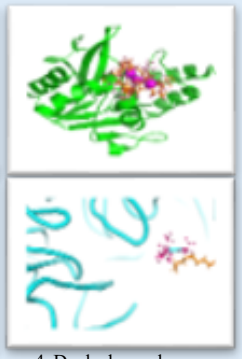
## COMPARISON TABLE

| Comparison | p-value | Significance |
|---|---|---|
| Exp v/s PC-1 | 0.002 | Highly |
| Exp v/s PC-2 | 0.00001 | Extremely |
| Exp v/s PC-3 | 0.00007 | Extremely |

1. Retrieved protein



2. Active site is marked

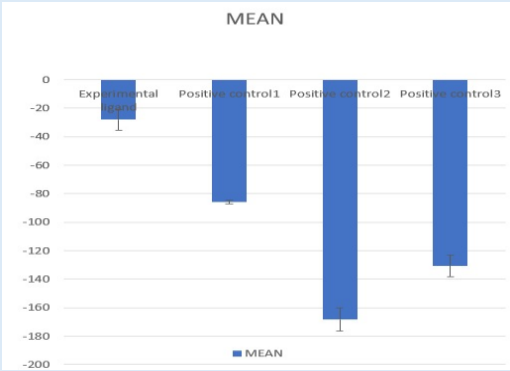




3. Experimental drug retrieved from pubchem



4. Docked complex




5. Graph plotted between AC score vs Mean ± S.D.

## METHODOLOGY

Compounds are retrieved from pubchem (TRIENTINE, PUTRESCINE, SPERMIDINE, SPERMINE) → Target protein retrieved form PDB (PDB ID: 6KZL) → Preparation of target protein in pymol → Molecular docking (swissdock) → Protein ligand interaction (PLIP) analysis of H-bonding. → Statistical analysis using MS Excel (Mean ± S.D. and T-test with unequal variance) → ADME prediction using swissADME

## RESULTS

- This study demonstrates that Zinc chelating drugs can interact with the active site residues of NDM-1-beta lactamase suggesting their pharmacological inhibition towards the enzyme.
- Trientine is comparatively better drug inhibitor of NDM-1-beta lactamase and has more favorable pharmacokinetic properties than positive controls.
- ADME prediction indicates that Trientine has high solubility but, low gastrointestinal absorption and zero blood brain barrier penetration.
- It has minimal cytochrome P450 risk. The repurposed drug has safe metabolic profile but, limited oral absorption rate.

## CONCLUSION

This study highlights trientine as promising $Zn^{2+}$-chelating candidates for inhibiting NDM-1 and combating antibiotic resistance.

## REFERENCES

1. BD 2021 Antimicrobial Resistance Collaborators. (2024). Global burden of bacterial antimicrobial resistance 1990–2021: A systematic analysis with forecasts to 2050. The Lancet, 404(10459), 1199–1226. https://doi.org/10.1016/S0140-6736(24)01867-11.
2. Bahr, G., Gonzalez, L., & Vila, A. J. (2022). Metallo-β-lactamases and a tug-of-war for the available zinc at the host–pathogen interface. Current Opinion in Chemical Biology, 66, 102103. https://doi.org/10.1016/j.cbpa.2021.102103
3. Chang, H., Qiao, R., & Li, C. (2025). Recent advances in functionalized macrocyclic polyamines for medicine applications. Chinese Chemical Letters, 36(7), 110675. https://doi.org/10.1016/j.cclet.2024.110675

**KEYWORDS:** NDM-1 Enzyme, $Zn^{2+}$ Coordination, Metallo-β-lactamase Inhibitors, Antimicrobial Resistance

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